

Proposed Title

SentinelFlow: A Deep Learning-Based Intrusion Detection Framework Using Fast Fourier Transform-Enhanced Traffic Profiling for Big Data Network Security

Short Problem Statement

Existing deep learning-based **Intrusion Detection Systems (IDS)** often rely mainly on time-domain flow features and may miss hidden burst patterns, repeated traffic rhythms, and frequency-based attack signatures. **SentinelFlow** addresses this by using **Fast Fourier Transform (FFT)**-enhanced traffic profiling to improve deep learning-based intrusion detection on large-scale network traffic datasets.

Main Research Question

Can Fast Fourier Transform-enhanced traffic profiling improve the performance of deep learning models in detecting network intrusions from large-scale NetFlow datasets?

Specific Research Questions

1. **How can large-scale NetFlow traffic be transformed into time-windowed traffic signals suitable for FFT-based feature extraction?**
 2. **What frequency-domain features can be extracted from network traffic signals such as packet count, byte count, flow count, throughput, and inter-arrival time?**
 3. **Does adding FFT-based frequency-domain features improve the performance of deep learning models compared with using time-domain NetFlow features alone?**
 4. **Which deep learning model performs best for intrusion detection using SentinelFlow: Convolutional Neural Network, Long Short-Term Memory, Gated Recurrent Unit, or Transformer Encoder?**
 5. **Can FFT-enhanced features improve the detection of bursty, repetitive, or time-dependent attacks such as Distributed Denial-of-Service, Denial-of-Service, brute-force, botnet, and port-scanning attacks?**
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General Objective

To develop and evaluate **SentinelFlow**, a deep learning-based intrusion detection framework that uses **FFT-enhanced traffic profiling** to improve the classification of benign and malicious network traffic in large-scale NetFlow datasets.

Specific Objectives

1. To preprocess and clean large-scale NetFlow-based intrusion detection datasets.
 2. To transform raw flow records into time-windowed traffic signals.
 3. To extract FFT-based frequency-domain features from traffic signals.
 4. To create two feature sets: one using only time-domain features and another using time-domain plus FFT-enhanced features.
 5. To train and compare deep learning models, including **CNN**, **LSTM**, **GRU**, and **Transformer Encoder**.
 6. To evaluate model performance using accuracy, precision, recall, F1-score, macro F1-score, false positive rate, false negative rate, and confusion matrix.
 7. To determine whether FFT-enhanced features improve intrusion detection performance, especially for bursty and time-dependent attacks.
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Proposed Methodology

Phase 1: Dataset Preparation

The study will use a large-scale NetFlow-based intrusion detection dataset such as **NF-UQ-NIDS-v2**, **NF-CSE-CIC-IDS2018-v3**, or another NetFlow v3 dataset. The dataset should contain flow-level traffic features such as timestamps, packet counts, byte counts, throughput values, duration values, protocol information, and attack labels.

The dataset will first be stored in a structured project environment using a Dockerized Jupyter setup. A separate database preparation notebook will convert the raw dataset into efficient formats such as **Parquet** and optionally **DuckDB** for faster processing.

Phase 2: Data Cleaning and Exploratory Data Analysis

The dataset will be cleaned by handling missing values, infinite values, invalid timestamps, duplicate rows, and inconsistent labels. The class distribution will be analyzed to identify imbalance between benign and attack traffic.

Exploratory Data Analysis will examine:

- benign versus attack distribution
- attack type distribution
- packet and byte statistics
- flow duration behavior
- throughput behavior
- protocol and port patterns
- possible class imbalance issues

Phase 3: Time-Window Traffic Signal Construction

Network flow records will be grouped into fixed time windows, such as 1-second, 5-second, or 10-second windows. For each window, traffic signals will be created.

Examples of traffic signals include:

flows per window

packets per window

bytes per window

average flow duration

average throughput

retransmitted packets

unique source IP count

unique destination IP count

unique destination port count

This step is important because FFT should be applied to a sequence over time, not to a single independent row.

Phase 4: FFT-Based Feature Extraction

After constructing time-windowed signals, **Fast Fourier Transform** will be applied to each traffic signal. FFT converts traffic behavior from the time domain into the frequency domain.

The extracted frequency-domain features may include:

spectral energy

dominant frequency

spectral entropy

mean magnitude

maximum magnitude

low-frequency power

high-frequency power

high-to-low power ratio

burstiness score

periodicity score

These features are intended to capture repeated bursts, traffic rhythms, and abnormal frequency signatures that may appear during network attacks.

Phase 5: Feature Set Construction

Two main datasets will be created.

Feature Set A: Baseline Time-Domain Features

This contains the original NetFlow-derived features and statistical time-window features.

Feature Set B: FFT-Enhanced Features

This contains the original time-domain features plus FFT-based frequency-domain features.

The comparison between these two feature sets will answer whether FFT improves intrusion detection.

Phase 6: Deep Learning Model Training

Several deep learning models will be trained and compared.

Recommended models:

Model	Purpose
Multilayer Perceptron (MLP)	Basic deep learning baseline
Convolutional Neural Network (CNN)	Learns local feature patterns
Long Short-Term Memory (LSTM)	Learns sequential traffic behavior
Gated Recurrent Unit (GRU)	Alternative recurrent model with fewer parameters
Transformer Encoder	Learns broader dependencies across traffic features or sequences

Each model will be trained twice:

Model using time-domain features only

Model using time-domain + FFT-enhanced features

Phase 7: Model Evaluation

The models will be evaluated using:

accuracy

precision

recall

F1-score

macro F1-score

false positive rate

false negative rate

confusion matrix

training time

inference time

For intrusion detection, **recall**, **macro F1-score**, and **false negative rate** are especially important because missing an actual attack is more dangerous than simply achieving high accuracy.

Phase 8: Results Presentation

The final results will be presented using:

- model comparison tables
- confusion matrices
- baseline versus FFT-enhanced comparison
- attack class performance analysis
- HTML dashboard
- research discussion and conclusion

This will show whether SentinelFlow provides measurable improvement over time-domain-only deep learning models.

Expected Results

The study is expected to produce a working deep learning-based intrusion detection framework that can process large-scale NetFlow data, generate FFT-enhanced features, train multiple deep learning models, and compare their performance.

The expected technical outputs are:

1. A cleaned NetFlow intrusion detection dataset.
2. A time-windowed traffic signal dataset.
3. An FFT-enhanced feature dataset.
4. Trained deep learning models using baseline and FFT-enhanced features.

5. Model comparison results showing accuracy, precision, recall, F1-score, and macro F1-score.
6. A results dashboard presenting the performance of SentinelFlow.
7. A discussion identifying whether FFT-enhanced profiling improved intrusion detection.

The expected research outcome is that FFT-enhanced traffic profiling may improve the detection of bursty, repetitive, and time-dependent attacks such as **DDoS**, **DoS**, brute-force, botnet, and port-scanning attacks. However, the study will not assume improvement automatically. If FFT does not improve performance, that result will still be meaningful because it will show the conditions under which frequency-domain features are useful or not useful for deep learning-based intrusion detection.

Expected Contribution

The main contribution of SentinelFlow is its investigation of **frequency-domain traffic profiling** as an enhancement to deep learning-based intrusion detection. Instead of relying only on conventional time-domain NetFlow features, the study introduces FFT-derived spectral features and evaluates whether these features improve the performance of CNN, LSTM, GRU, and Transformer-based models.

In simpler terms:

SentinelFlow does not only ask whether deep learning can detect attacks. It asks whether deep learning can detect attacks better when network traffic is represented using both time-domain and frequency-domain behavior.

Acronym Glossary

IDS means **Intrusion Detection System**.

FFT means **Fast Fourier Transform**.

NetFlow means summarized network flow records.

CNN means **Convolutional Neural Network**.

LSTM means **Long Short-Term Memory**.

GRU means **Gated Recurrent Unit**.

MLP means **Multilayer Perceptron**.

DDoS means **Distributed Denial-of-Service**.

DoS means **Denial-of-Service**.

F1-score is a metric that balances precision and recall.